QUANT

# Venn Diagrams

eBOOK

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# INDEX

# Venn Diagram E- Book


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# SETS & Venn Diagram

## 1.1 Set Theory

A set is well–defined collection of objects. Sets are usually denoted by A, B, C, ——. The objects in the set are called 'elements' of the set.
If $x$ is an element of the set A, we say, $x \in$ A ($x$ belongs to A).
Sets are usually defined in two forms – the tabular form and the roster method.

**Tabular form:** In the tabular form, all elements of the set are enumerated or listed. e.g. The set of natural numbers: N = {1, 2, 3, 4, – –}, the set of integers: I = {– –, –3, – 2, –1, 0, 1, 2, 3, – –}.

**Roster method:** Under this method, the defining property of the set is specified. If all the elements in the set have a property P, then we can define the set as A = {$x$ : $x$ has the property P}.
E.g. N = {$x$ : $x$ is a natural number}, I = {$y$ : $y$ is an integer}.

## 1.2 Types of Sets

**Finite sets:** A set with a finite number of elements is called a finite set. e.g. A = {$a$, $e$, $i$, $o$, $u$}.

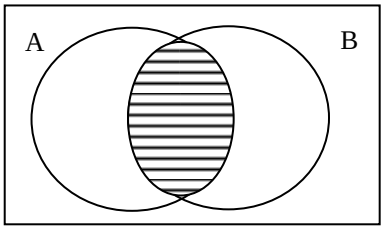
**Empty set:** A set which contains no elements is called an empty set or a null set. An empty set is denoted as '{}' or $\phi$.

**Infinite set:** A set that is neither an empty set nor a finite set is called an infinite set. Such a set will contain infinitely many elements. e.g. $N$ = {1, 2, 3, —}.

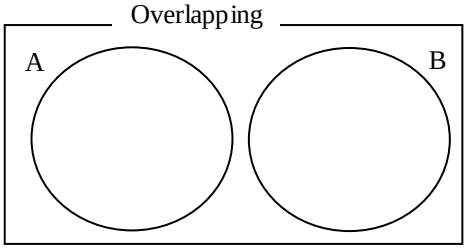
**Disjoint sets:** Two sets are said to be disjoint sets if they do not have any common elements. e.g. If A = {$x$ : $x$ is an even number} and B = {$y$ : $y$ is an odd number}, then A and B will be disjoint sets.

**Overlapping sets:** If two sets A and B have some common elements, the sets are said to be overlapping. e.g. A = {1, 2, 3, 4, 5} and B = {3, 4, 5, 6, 7}. These sets are overlapping sets as the elements 3, 4 and 5 are common to both sets. The elements common to both sets is called the intersection of the two sets and is denoted as A $\cap$ B. On the other hand, the addition of two sets is called the union of two sets and is denoted by A $\cup$ B.

**Equal sets:** Two sets are said to be equal if they contain the same elements. e.g. A = {2, 4, 6, 8}




Overlapping




Non – overlapping

B = {x : x is an even number between 1 and 9} are equal sets as they contain the same elements.

**Universal set:** A universal set is the set that contains the elements of all the sets under consideration. It is usually denoted by U, e.g. A = {4, 7, 8, 9}, B = {–4, –2, 0, 1, 4, 7, 10}. The set of integers, I = {– –, –2, –1, 0, 1, 2, – –} will be the universal set for A and B.

**Complement of a set:** Given a set A, the complement of a set is the set that contains elements not belonging to A and is denoted by Ac. The union of the given set and its complement will give the universal set i.e. U = A + Ac. e.g.
A = {x : x is a Mathematics book in the Bulls Eye library}.
Ac = {y : y is not a Mathematics book in the Bulls Eye library} And the universal set in this case will be
U = the set of all books in the Bulls Eye library.

## 1.3 Basic Operations on Sets

1. **Intersection of two sets:** The intersection of two sets is the set of elements common to both the given sets.
   The intersection of two sets A and B is denoted as $A \cap B$.
   In notation form, we can define the intersection of two sets A and B as $A \cap B = \{x : x \in A, x \in B\}$.
   If, $A \cap B = \phi$, then A and B are disjoint sets.
   If $A \cap B \neq \phi$, then A and B are overlapping sets.

2. **Union of two sets:** The union of two sets is the set containing the elements belonging to A and also the elements belonging to B. The union of these sets is denoted by $A \cup B$. In notation form, we can define the union of two sets as $A \cup B = \{x : x \in A, x \in B, x \in A \cap B\}$.

3. **Difference of two sets:** The difference of two sets A and B is the set of elements that belong to A but do not belong to B.
   The difference of two sets is denoted by A – B. In notation form, we can define the difference of two sets as $A - B = \{x : x \in A, x \notin B\}$.

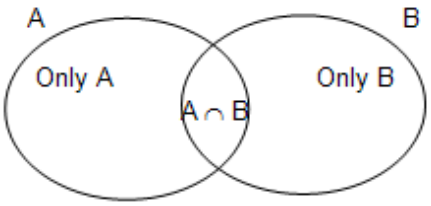
| SYMBOL | MEANING | EXAMPLE |
|---|---|---|
| { } | is a set | $S = \{4,5\}$ |
| $\in$ | is an element of | $s \in S$ |
| $\notin$ | is not an element of | $s \notin T$ |
| $\subseteq$ | is a subset of | $S \subseteq T$ |
| $\subset$ | is a proper subset of | $S \subset T$ |
| $\cup$ | union | $S \cup T$ |
| $\cap$ | intersection | $S \cap T$ |
| $\varnothing$ | the empty set | $\{2,3,4\} \cap \{5,6,7\} = \varnothing$ |

## 1.4 Venn Diagram

**Venn Diagram:** A Venn diagram is a closed figure used to denote the set of all points within the figure.

**Venn Diagram of two objects:** If the two given objects are called ‘A’ & ‘B’. The sum of ‘A’ & ‘B’ here is known as “A union B” and is symbolically represented as “$A \cup B$”.
$A \cup B = A + B - A \cap B$.

Where “$A \cap B$” $\Rightarrow$ “A intersection B” $\Rightarrow$ that part which is common in A & B. “$A \cap B$” is subtracted once from the sum of A & B because it is included twice in the total, being a part of both A & B.




If the value of 'A union B' is to be calculated from the diagram, the sum of these 3 values given inside the diagram will give you "$A \cup B$". Because one part represents only 'A', other part only 'B' and the third common part "$A \cap B$".

## 1.5 Solved Examples

**Ex.1** In a locality, 2500 persons read Times of India and 3500 persons read The Tribune. There are 250 persons who read both of these newspapers. Find the number of persons who read 1 or 2 of these newspapers.

**Sol:** It is very much clear after reading the question that we have to find the value of '$A \cup B$'.
$\Rightarrow A \cup B = 2500 + 3500 - 250 = 5750$.

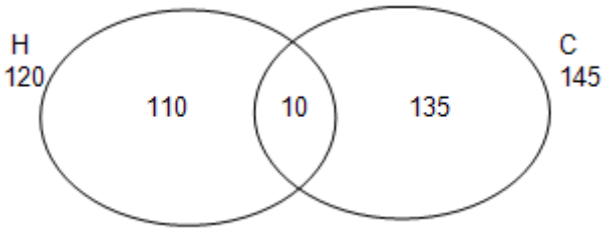
**Ex.2** In a class of 64 students, 50 % of the students have taken Sociology and 75 % of the students have taken Politics.
How many students have taken both the subjects?

**Sol:** We have $S = 64 \times \frac{50}{100} = 32$, $P = 64 \times \frac{75}{100} = 48$.
Now $(S \cup P) = S + P - (S \cap P)$
$\Rightarrow 64 = 32 + 48 - (S \cap P)$.
Therefore, $(S \cap P) = 80 - 64 = 16$.

**Ex.3** In a class, 120 students play Hockey, 145 students play Cricket. There are 10 students who play both of these games. The total number of students in the class is 280. What is

a) The number of students playing 1 or more of games?
b) The number of students playing no game?
c) The number of students playing only Cricket?
d) The number of students playing only Hockey?

**Sol:** As we have to answer four questions, it is better to make diagram in this case.

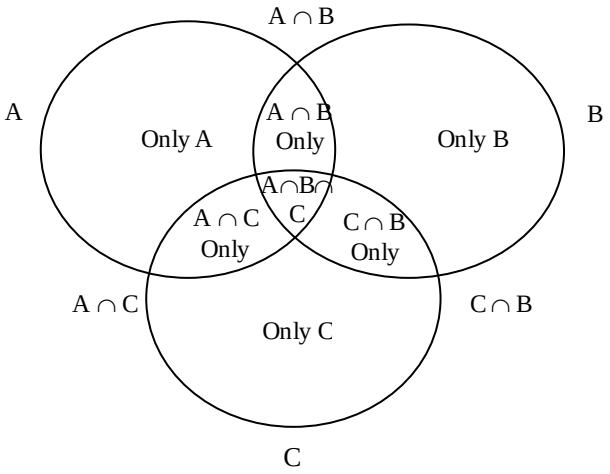



a) The students who play one or more of these two games means the value of $H \cup C \Rightarrow 110 + 10 + 135 = 255$.
b) The total number of students in the class is given to be 280, as only 255 play one or more games, this implies the remaining students play none of these games i.e. $280 - 255 = 25$.
c) It can be seen from the diagram that there are 135 students who play only Cricket.
d) It can be seen from the diagram that there are 110 students who play only Hockey.

## 1.6 Venn Diagram of 3 objects

If the three given objects are called 'A', 'B' & 'C', the sum of 'A', 'B' & 'C' is known as "A union B union C" & is represented as "$A \cup B \cup C$".

$A \cup B \cup C = A + B + C - A \cap B - B \cap C - C \cap A + A \cap B \cap C$.




If "$A \cup B \cup C$" is to be calculated from the diagram, then all the 7 parts given inside the circle should be added i.e. only A, only B, only C, only $A \cap B$, only $B \cap C$, only $C \cap A$ and $A \cap B \cap C$.

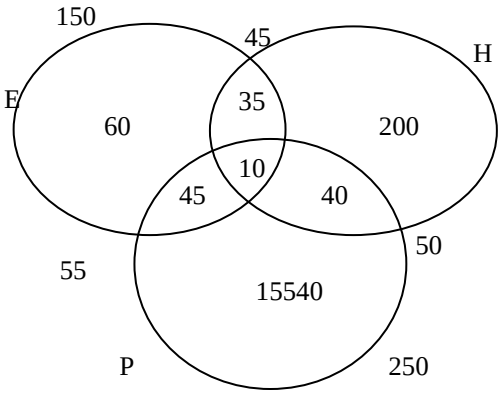
It is very important here to understand the meaning of certain terms.

**<u>At least 1</u>:** means minimum 1 i.e. 1 or more than 1.
**<u>At least 2</u>:** means minimum 2 i.e. 2 or more than 2.
**<u>At the most 2</u>:** means maximum 2 i.e. 2 or less than 2.
**<u>At the most 3</u>:** means maximum 3 i.e. 3 or less than 3.

**Ex.4** In a school 150 students have English, 200 have Hindi only, 250 have Punjabi. 45 students have Hindi & English, 50 students have Hindi & Punjabi, 55 students have Punjabi & English. There are 10 students, who have all the 3 languages. Having at least one of these languages is a must. Calculate the number of students

1. In the school.
2. Having only English.
3. Having only Hindi & English.
4. Having exactly 1 language.
5. Having exactly 2 languages.
6. Having at least 1 language.
7. Having at least 2 languages.
8. Having at most 3 languages.
9. Having Hindi & one other language.
10. Having English & one or more other languages.

**Sol:** First of all the diagram will be made. Take care in this case, that the second sentence specifies that 200 have Hindi only, means it is to be written inside the Hindi circle.



1. Number of students in school = 60 + 200 + 155 + 35 + 45 + 40 + 10 = 545.
2. Only English = 60 (diagram).
3. Only Hindi & English = 35 (common part between H & E)
4. Exactly 1 language = 60 + 200 + 155 = 415.
5. Exactly 2 languages = 35 + 45 + 40 = 120.
6. 1 or more than 1 $\Rightarrow$ includes all 3 = 545.
7. 2 or more than 2 $\Rightarrow$ 35 + 45 + 40 + 10 = 130.
8. 3 or less than 3 $\Rightarrow$ includes all 3 = 545.
9. 40 + 35 = 75.
10. 35 + 45 + 10 = 90.

**Ex. 5** Of the 400 members of a sport club, 144 play Squash (S), 135 play Badminton (B), 156 play Table Tennis (T). Each member plays at least one game. The members who play badminton and table tennis both must play squash also. For every three members who play at least two games, there are two members who play all the three games. Find the number of members who play all the three games.

**Sol:** Given S = 144, B = 135 and T = 156. Let $x$ members play all the three games. Refer Venn Diagram, given below.

As per the question, $\frac{a+b+x}{3} = \frac{x}{2}$

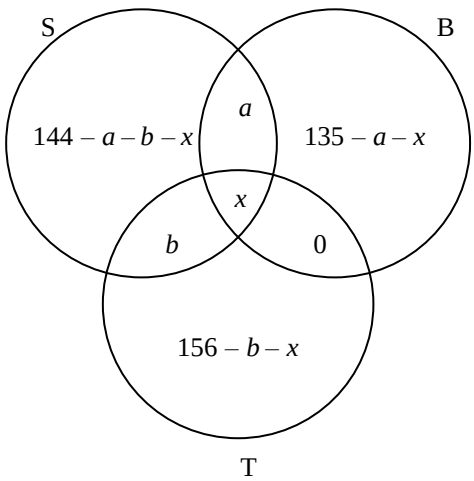
$\Rightarrow x = 2(a + b)$ or $a + b = \frac{x}{2}$.

Now adding all these values and equating it to 400, we get $144 - a - b - x + a + 135 - a - x + b + x + 156 - b - x = 400$.

$\Rightarrow 435 - 2x - (a + b) = 400$.

So $435 - 2x - \frac{x}{2} = 400 \Rightarrow x = 14$.

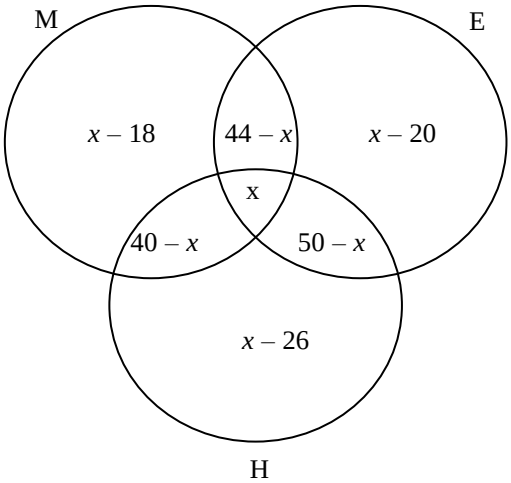
Hence 14 members play all the three games.




**Ex.6** A study of 104 students, who speak Marathi or English or Hindi, reveals 38 do not speak Marathi, 30 do not speak English and 40 do not speak Hindi. 40 speak Hindi and Marathi, 50 speak Hindi and English, 44 speak Marathi and English.

a) How many speak all three languages?
b) How many speak Hindi but neither Marathi nor English?
c) How many speak exactly two languages?

**Sol:** Total = 104 students. If 38 students do not speak Marathi, so students speaking Marathi = 104 – 38 = 66. Similarly, students speaking English = 104 – 30 = 74 and students speaking Hindi = 104 – 40 = 64. Now drawing the Venn diagram,




Adding all these values and equating it to 104, we get $x - 18 + 44 - x + x - 20 + 40 - x + x + 50 - x + x - 26 = 104$. So $x = 34$.
So number of students speaking Hindi but not Marathi or English = $x - 26 = 8$. Number of students speaking exactly 2 languages = $44 - x + 40 - x + 50 - x = 44 - 34 + 40 - 34 + 50 - 34 = 32$.

## 1.7 Venn diagram with 4 variables

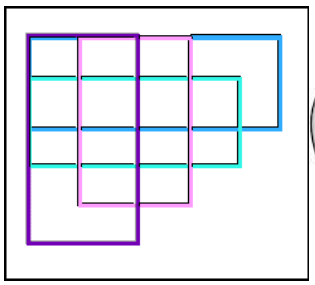
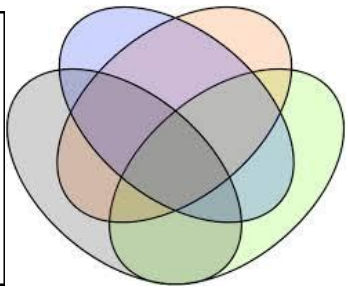
Let there be 4 sets (A, B, C & D). For each object, there are two possibilities:
1) either it belongs to a particular set or
2) not.
This would give us a total of 2 × 2 × 2 × 2 = 16 possibilities.

**Given below are all possible intersections of the four sets:**

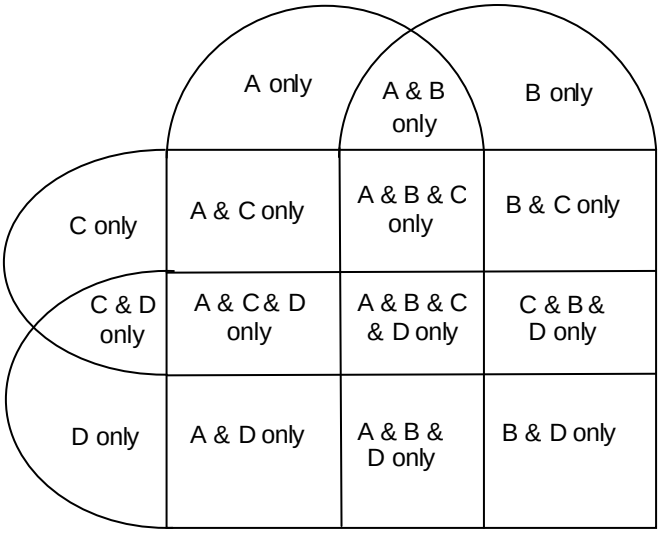
| | | | |
|---|---|---|---|
| 1. {A, B, C, D} | 2. {A, B, C} | 3. {A, B, D} | 4. {A, B} |
| 5. {A, C, D} | 6. {A, C} | 7. {A, D} | 8. {A} |
| 9. {B, C, D} | 10. {B, C} | 11. {B, D} | 12. {B} |
| 13. {C, D} | 14. {C} | 15. {D} | 16. { } |

One of these possibilities is when no object is a part of any of the sets i.e. empty { }.
Hence, the Venn - Diagram that we draw will need 16 - 1 = 15 separate sections.
Now, it is impossible to divide a plane into the 16segments required for a complete 4-set Venn diagram using only circles as we could do for less than 4 objects because two circles intersect in at most two points, and each intersection creates one new region. Since the fourth circle intersects the first three in at most 6 places, it creates at most 6 new regions; that's 14 total, but we need 16 regions to represent all possible relationships between four sets. Yet, it is doable using ellipses or rectangles as two ellipses /rectangles can intersect at more than two points creating more regions.

Given below are few diagrams of a 4-set Venn diagram.

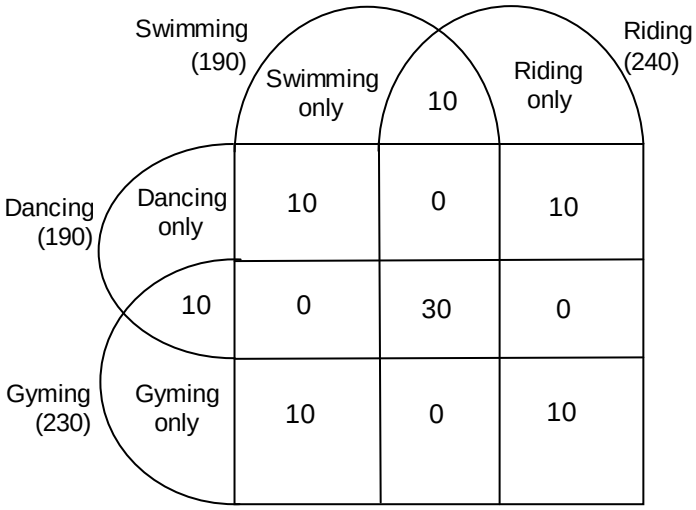
But the most convenient way to represent this is shown below. It clearly represents each of the 15 sections separately.




**Ex.7** In a community, members indulge in four different activities–Riding, Swimming, Dancing and Gyming. All persons indulge in at least one of the activities but 20% of them do nothing. 240, 190, 190 and 230 persons carry out the four activities given in the above order respectively. The number of persons involved in exactly 2 activities, for any two activities is 10. There are 30 persons who do all the four activities, but there is nobody who does exactly three out of four activities.

Answer the following questions**.**
**Question 1:** How many persons do exactly one activity?
**Question 2:** How many persons do exactly two activities?
**Question 3:** How many persons do nothing at all?
**Question 4:** What percentage of the people doing swimming also do at least one other activity?

**Sol:** Let us draw the Venn – Diagram using the information given.




It is given that; the number of people involved in exactly 2 activities, for any two activities are 10. This means that each portion that represents exactly two activities is 10.
We should now try to fill the missing areas, starting with the number of people involved in exactly one activity.
People doing only Riding: 240 - (10+10+10+30) = 240 - 60 = 180
People doing only Swimming: 190 - (10+10+10+30) = 190 - 60 = 130
People doing only Dancing: 190 - (10+10+10+30) = 190 - 60 = 130
People doing only Gyming: 230 - (10+10+10+30) = 230 - 60 = 170

**Let us solve the questions now:**

**Sol 1:** As we have calculated, the number of people involved in exactly one activity are =180+130+130+170= **610 persons.**

**Sol 2:** Refer to the diagram above. There are 6 portions representing exactly two ($^4C_2$= 6). Each portion is 10 as given in the question. Therefore people involved in exactly two activities are= 6×10 = **60 persons.**

**Sol 3:** We know that 20% of the people do nothing. This means that 80% of the people do at least one activity. These 80% people are represented in the 15 regions in the above Venn - Diagram. So,80% of total = people involved in exactly one activity + in exactly 2 activities + exactly 3 activities + exactly 4 activities = (180+130+130+170)+6×10+0+ 30= 700. Total = 700/0.8 = 875.People who do nothing = 20% of 875 =**175 persons.**

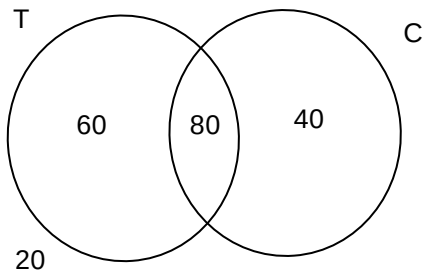
**Sol 4:** Out of the 190 people who do Swimming, 130 do only Swimming. This means that the other people which is equal to (190-130)=60, also do at least one other activity. Required percentage = (60/190) × 100 = **31.57%**

## 1.8 Solved Examples:

**Ex.1** In a college, 200 students are randomly selected. 140 like tea, 120 like coffee and 80 like both tea and coffee.

1. How many students like only tea?
2. How many students like only coffee?
3. How many students like neither tea nor coffee?
4. How many students like only one of tea or coffee?
5. How many students like at least one of the beverages?

**Sol:** The given information may be represented by the following Venn diagram, where T = tea and C = coffee.



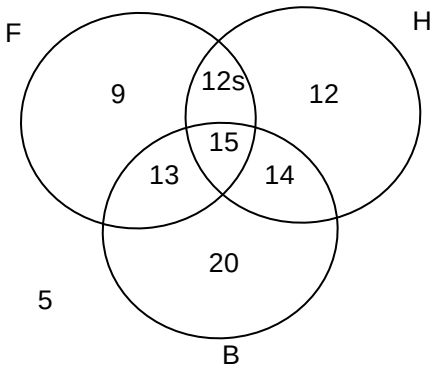

1. Number of students who like only tea = 60
2. Number of students who like only coffee = 40
3. Number of students who like neither tea nor coffee = 20
4. Number of students who like only one of tea or coffee = 60 + 40 = 100
5. Number of students who like at least one of tea or coffee = n ( only Tea )+ n ( only coffee ) + n ( both Tea & coffee ) = 60 + 40 + 80 = 180

**Ex. 2** In a survey of 500 students of a college, it was found that 49% liked watching football, 53% liked watching hockey and 62% liked watching basketball. Also, 27% liked watching football and hockey both, 29% liked watching basketball and hockey both and 28% liked watching football and basket ball both. 5% liked watching none of these games.

1. How many students like watching all the three games?
2. Find the ratio of number of students who like watching only football to those who like watching only hockey.
3. Find the number of students who like watching only one of the three given games.
4. Find the number of students who like watching at least two of the given games.

**Sol:** n(F) = percentage of students who like watching football = 49%
n(H) = percentage of students who like watching hockey = 53%
n(B)= percentage of students who like watching basketball = 62%
n ( F ∩H) = 27% ; n (B ∩H) = 29% ; n(F ∩B) = 28%
Since 5% like watching none of the given games so, n (F U H U B) = 95%.
Now applying the basic formula,
95% = 49% + 53% + 62% -27% – 29% – 28% + n (F ∩H ∩B)
Solving, you get n (F ∩H ∩B) = 15%.
Now, make the Venn diagram as per the information given.

**Note:** All values in the Venn diagram are in percentage.




1. Number of students who like watching all the three games = 15 % of 500 = 75.
2. Ratio of the number of students who like only football to those who like only hockey = (9% of 500)/(12% of 500) = 9/12 = 3:4.
3. The number of students who like watching only one of the three given games = (9% + 12% + 20%) of 500 = 205
4. The number of students who like watching at least two of the given games = (number of students who like watching only two of the games) +(number of students who like watching all the three games) = (12 + 13 + 14 + 15)% i.e. 54% of 500 = 270.

**Ex.3** A school has 65 students studying English, Maths and Physics. 31 study English, 22 Maths and 24 Physics. 10 study English and Maths, 8 study Physics and Maths while 6 study both English and Physics. How many study all the three subjects?

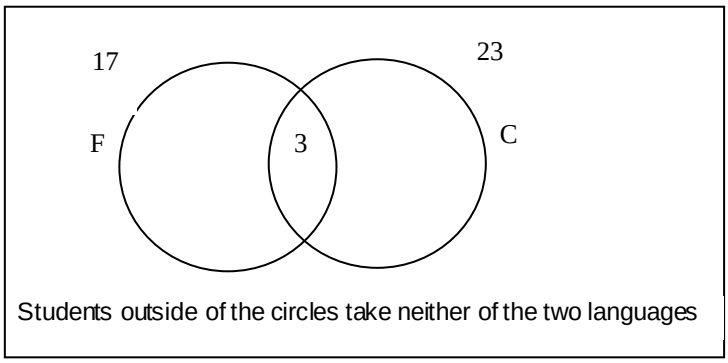
**Sol:** Let A, B and C denotes the no of students studying English, Maths and Physics respectively.
No of students studying all the three subjects = n(A ∩ B ∩ C)
We know,
n(A U B U C) = n(A) + n(B) + n(C) – n(A ∩ B) - n(B ∩ C) - n(A ∩ C) + n(A ∩ B ∩ C)
65 = 31 + 22 + 24 -10 – 8 - 6 + n(A ∩ B ∩ C)
n(A ∩ B ∩ C) = 65 – 53 = 12.

**Ex.4** There are 120 students at DAV Public School. 72 students play football and 57 play hockey. How many of these students play both football and hockey?

**Sol:** Here, No of students who play football n(A) = 72
No of students who play hockey n(B) = 57
So, the no of students who play both football and hockey = $n(A \cap B)$
We know, $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
$\Rightarrow 120 = 72 + 57 - n(A \cap B)$
$\Rightarrow n(A \cap B) = 129 - 120 = 9$

**Ex.5:** In a class of 60 students, 17 opt for French, 23 opt for Chinese, and 3 opt for both French and Chinese. How many students in the class are not enrolled in either of the two languages French or Chinese?

**Sol:** 60




Here, No of students who take French n(A) = 17
No of students who take Chinese n(B) = 23
So, the no of students who take both French and Chinese = $n(A \cap B) = 3$
Now, No of students who take neither of the two languages = 60 – [17 + 23 - 3] = 60 – 37 = 23.

**Ex.6** In Delhi Public School, 45 students study Biology and English both. If 60 students study biology, and 75 students study English. How many of these students study at least one of the two subjects?

**Sol:** Here, No of students who study Biology n(A) = 60
No of students who study English n(B) = 75
So, the no of students who study both Biology and English = $n(A \cap B) = 45$
Now, no. of students who study at least one of the two subjects = $n(A \cup B)$
So, $n(A \cup B) = n(A) + n(B) - n(A \cap B) = 60 + 75 - 45 = 90$.

**Ex.7** In a school of 420 students, there are two clubs, dramatics club and dance club. 180 students are in dramatics club, 70 students are the members of both the clubs. How many students are involved in dance club?

**Sol:** Here, No of students who participate in at least one of the two subjects $n(A \cup B) = 420$
No of students who are in dramatics club n(A) =180
No of students who participate in both the clubs
$n(A \cap B) = 70$
Now, No of students who are in dance club = n( B)
Now, $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
$420 = 180 + B - 70 \Rightarrow n(B) = 310$

**Ex.8** In a class there are:

- 8 students who study both Maths and English.
- 7 students who studies none of Maths and English
- 13 students who study Maths
- 19 students who study English.

How many students are there in the class?

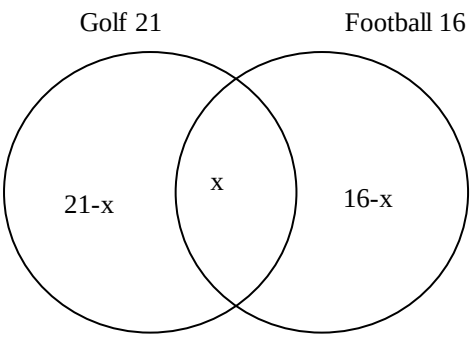
**Sol:** n(AUB) = n(A) + n(B) – n(A ∩B)
∴ n( AUB) = 19+13-8 = 24
In the universal set, The elements which are not included in AUB are also added, i.e. elements not being the part of AUB. Therefore total people are = 24+7 = 31.

**Ex.9** In a class there are 30 students.

- 21 students like Golf
- 16 students like Football.
- 6 students like neither Golf nor Football

How many students like both Golf and Football?

**Sol:**




AUB = Universal set – None of them
∴ n(AUB) = 30-6 = 24
Further n(AUB) = n(A) + n(B) – n(A∩B)
∴ 24 = (21-x) + (x) + (16-x)
Solving, we get x = 13
So we can conclude that 13 students play both the games.

**Ex.10** There are 120 students in the class, All even numbered students opt for Psychology, Those whose numbers are divisible by 5 opt for Sociology and those whose numbers are divisible by 7 opt for Biology. How many opt for none of the three subjects?

**Sol:** We need to find out the number of students who took at least one of the three subjects and subtract that number from the overall 120 to get the number of students who did not opt for any of the three subjects.

Number of students who took at least one of the three subjects can be found by finding out A U B U C, where A is the set of those who took Psychology, B the set of those who took Sociology and C the set of those who opted for Biology.
Now, n(AUBUC) = n(A) +n( B) +n(C) - n(A ∩ B) – n( B ∩ C) – n( C ∩ A) +n (A ∩ B ∩ C)
A is the set of those who opted for Psychology = 120/2 = 60 students
B is the set of those who opted for Sociology = 120/5 = 24
C is the set of those who opted for Biology = 120/7 = 17.

The 10th, 20th, 30th..... numbered students would have opted for both Psychology and Sociology.
Therefore, $n(A \cap B) = 120/10 = 12$
The 14th, 28th, 42nd..... Numbered students would have opted for Psychology and Biology.
Therefore, $n(C \cap A) = 120/14 = 8$
The 35th, 70th.... numbered students would have opted for Sociology and Biology.
Therefore, $n(B \cap C) = 120/35 = 3$
And the 70th numbered student would have opted for all three subjects.
Therefore, $n(A \cup B \cup C) = 60 + 24 + 17 - (12 + 8 + 3) + 1 = 79$.
Number of students who opted for none of the three subjects = 120 - 79 = 41.

**Ex.11** In the class,12 students enrolled for both Ballet dancing and Salsa. 22 enrolled for Salsa. If the students of the class enrolled for at least one of the two dance forms and there are 40 students in the class, then how many students enrolled for only Ballet dancing and not Salsa?

**Sol:** Let A be the set of students who have enrolled for Ballet dancing and B be the set of students who have enrolled for Salsa.
Then, $(A \cup B)$ is the set of students who have enrolled at least one of the two dance forms. As the students of the class have enrolled for at least one of the two dance forms, $n(A \cup B) = 40$
We know $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ i.e., $40 = n(A) + 22 - 12$ or $n(A) = 30$ which is the set of students who have enrolled for Ballet dancing and includes those who have enrolled for both the dance forms.
However, we need to find out the number of students who have enrolled for only Ballet dancing = Students enrolled for Ballet dancing - Students enrolled for both Salsa and Ballet dancing = 30 - 12 = 18.

**Ex.12** In a society, 40% of the people opted for Gaming and 70% opted for Studying. If 15% of the people opted for both Gaming and Studying, what % of the people of the society did not opt for either of the two activities?

**Sol:** We know that $n(A \cup B) = n(A) + n(B) - n(A \cap B)$,
Where $(A \cup B)$ represents the set of people who have opted for at least one of the two activities Gaming or Studying and $(A \cap B)$ represents the set of people who have opted for both the activities Gaming and Studying.
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
$\Rightarrow n(A \cup B) = 40 + 70 - 15 = 95\%$
That is 95% of the people have opted for at least one of the two activities Gaming or Studying.
Therefore, the balance (100 - 95)% = 5% of the people have not opted for either of the two activities.